

Project: Reservoir Computing for Connectome Networks I

Description

The rate model represents the activity of each neuron (in spikes per second) as a continuous variable. The rate of the i -th neuron at time t is represented by $r_i(t)$. The temporal evolution for N neurons follows the system of differential equations

$$\frac{dr_i}{dt} = -r_i + S \left[\sum_{j=1}^N W_{ij} r_j(t) \right],$$

where N is the number of neurons, $S(x) = \tanh(x)$ is a signmodal nonlinearity which relates the incoming current to the change in neuron's firing rate, and W_{ij} the recurrent connectivity matrix.

Using an Euler method, the differential equations can be integrated numerically to analyze the system's behavior for different network topologies. This involves plotting individual trajectories and analyzing the statistics of the rates. The initial condition is chosen randomly within a small interval.

Tasks

We want to implement several networks that will serve as control to compare our results to. We begin with the **Gaussian random network**, a fully connected network where all elements are independent Gaussian variables of mean zero and variance $\frac{J^2}{N}$. Thus, $W_{ij} = \mathcal{N}(0, \frac{J^2}{N})$. The task is to simulate the activity of the rates for different values of J . A sharp change should happen at a critical value $J_c = 1$: for $J < J_c$, rates will decay to zero at large times. For $J > J_c$, they will have display chaotic, continuously moving activity. A way to visualize this is to compute the variance of the time series of $r_i(t)$ over time; then, average the result over all N neurons. This quantity is larger than zero only for $J > J_c$.

In the future, we will also implement other architectures. For now, there is nothing to do with them:

1. Fully Connected Network

- All links have the same weight: $W_{ij} = \frac{J}{N}$

2. Erdős-Rényi Network

- Nodes are randomly connected with probability p .
- The number of connections is $k = pN$, which is the control parameter.
- The strength of the connections is homogeneous, $\frac{J}{k}$

3. Data Integration

- The network data is incorporated by setting $W_{ij} = JD_{ij}$ where D_{ij} represents the network data and J is an arbitrary scaling factor.

Some references

- A similar study showing an important error, as it does not control for network sizes. <https://arxiv.org/abs/2211.00161>.

- Some discussion on reservoir computing as a tool to understand the brain <https://doi.org/10.1098/rstb.2018.0377>. Another brief review <https://arxiv.org/html/2412.13212v1>
- A hands-on tutorial on how to build reservoirs (in Julia) <https://www.martinuzzi.net/posts/a-brief-introduction-to-reservoir-computing/>
- Connectomic dataset preprocessed: <https://braingraph.org/cms/>. On the HCP data collection, original publications, etc.: <https://www.humanconnectome.org/>